Silicon Nitride Integrated Photonic CNOT Gate For Linear Optical Quantum Computing

By Morgan McCabe

1. Abstract

This report presents the design of a proof-of-concept silicon nitride 4×4 unitary photonic CNOT gate at 780nm for linear optical quantum computing. The device includes grating couplers, MMIs, a thermo-optically tuned MZI, and an AMZI modeled using Interconnect. Coupling efficiency results from each individual component provide strong basis for future fabrication.

2. Introduction

Linear optical quantum computing uses interference, coupling, and phase control to realise quantum logic in integrated photonic circuits. This report presents a proof-of-concept silicon nitride 4×4 unitary CNOT architecture at 780 nm, designed within CORNERSTONE constraints and evaluated through simulation of grating couplers, MMIs, thermo-optic MZIs, and AMZI components.

3. Design Concept

The device is a proof-of-concept SiN 4×4 unitary photonic circuit for a linear optical CNOT gate at 780 nm, comprising grating couplers, 50:50 and 33:66 MMIs, an MZI, and an AMZI, with single-mode operation targeted in the fundamental TE mode. Together, these components form the complete photonic circuit required to realize the target linear optical CNOT operation.

4. Design Rationale

The foundry assigned for this project was CORNERSTONE UK, based in Southampton [2]. Consequently, the design decisions were constrained by both the fabrication capabilities of the foundry and the technical requirements needed to realize the intended device.

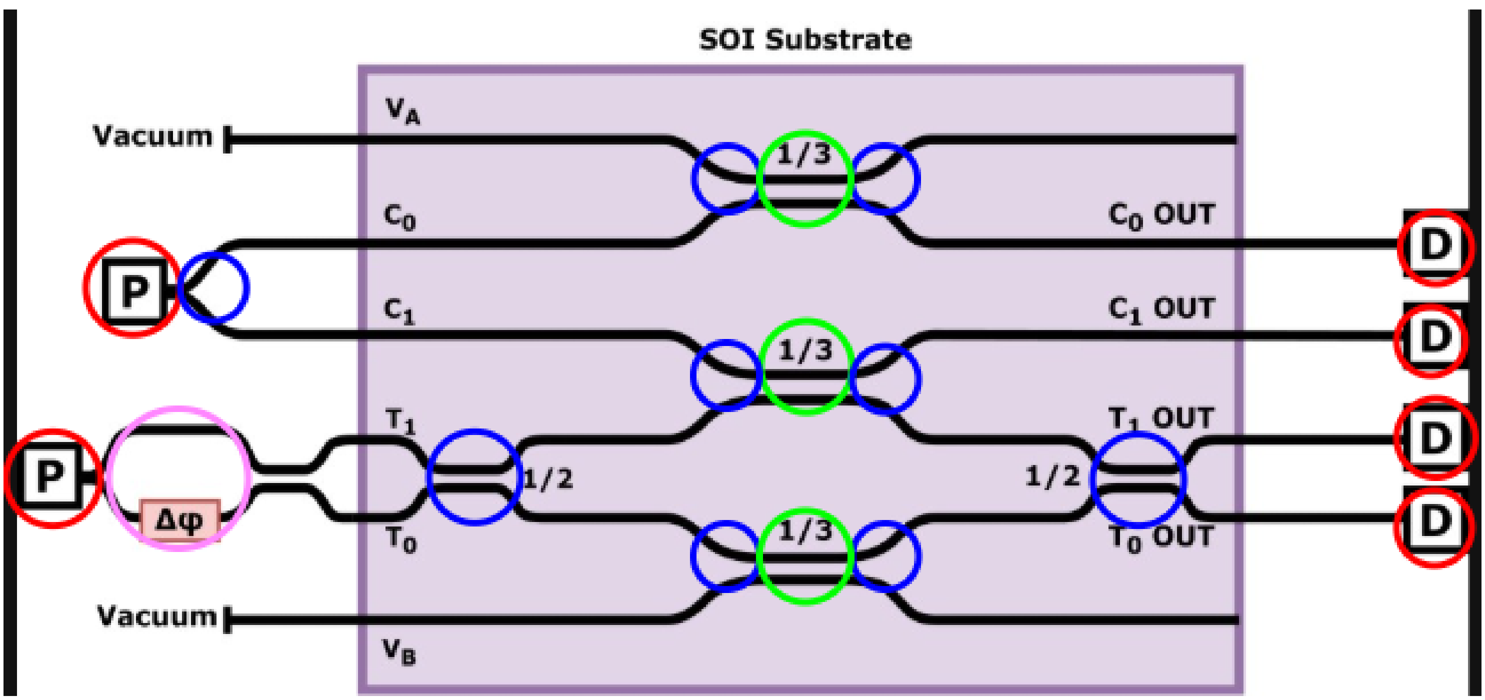

Fig. 4.1. 4x4 Unitary

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A **Grating couplers**: fibre–chip input/output coupling.

B **50:50 MMI couplers**: equal power splitting/recombination for interference.

C **33:66 MMI couplers**: unequal splitting for the required transformation probabilities.

D **MZIs**: phase control between optical paths.

E **AMZIs**: asymmetric path-length difference for additional phase control.

4.1.) General Values

4.1.1.) Wavelength

A wavelength of 780 nm was chosen for its relevance to visible-range quantum photonics, particularly rubidium-based quantum-memory systems [3]. It also allows investigation of a less mature integrated-photonics design space than the standard 1550 nm telecommunications regime [4].

4.1.2.) Material

The waveguide core material was chosen as Si_3N_4 due to its suitability for operation at 780 nm and its availability within the CORNERSTONE visible photonics platform [5; 6]. The top and bottom cladding layers were chosen as SiO_2, consistent with standard silica-based integrated photonic structures [7]. Together, these materials provide sufficient refractive-index contrast to confine the optical mode within the core. $\Delta n_{\text{eff}} \approx 0.209$

4.1.3.) Waveguide dimensions and guided modes

The waveguide height was set to 0.3 μm, consistent with the CORNERSTONE SiN platform specification [8]. The width was set to 0.35 μm to support single-mode operation at 780 nm, as verified by the FDE mode analysis (Figs. 3.1–3.6).

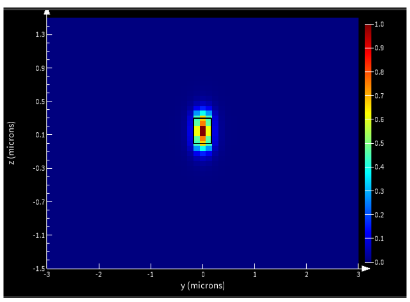


Fig. 4.2. 0.35 μm, TE Mode 1

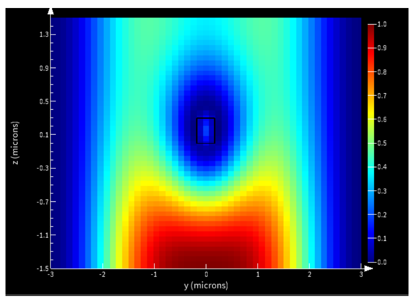


Fig. 4.3. 0.35 μm, TE Mode 2

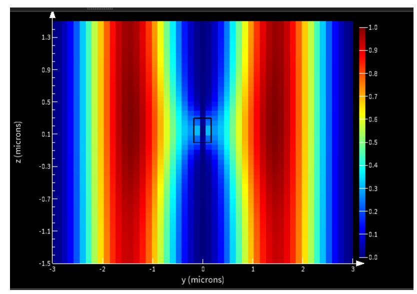


Fig. 4.4. 0.35 μm, TE Mode 3

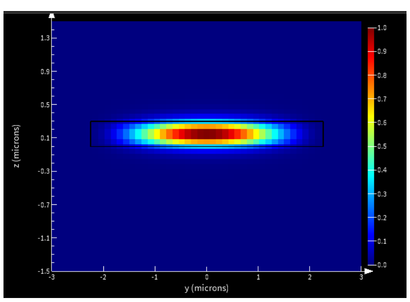


Fig. 4.5. 4.5 μm, TE Mode 1

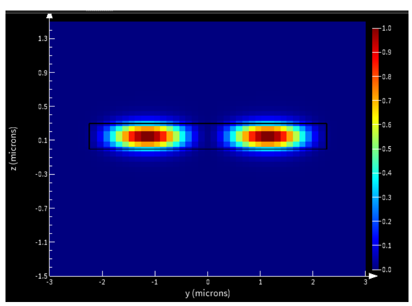


Fig. 4.6. 4.5 μm, TE Mode 2

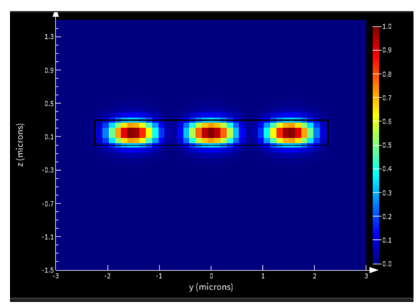


Fig. 4.7. 4.5 μm, TE Mode 3

4.2.) Grating Couplers

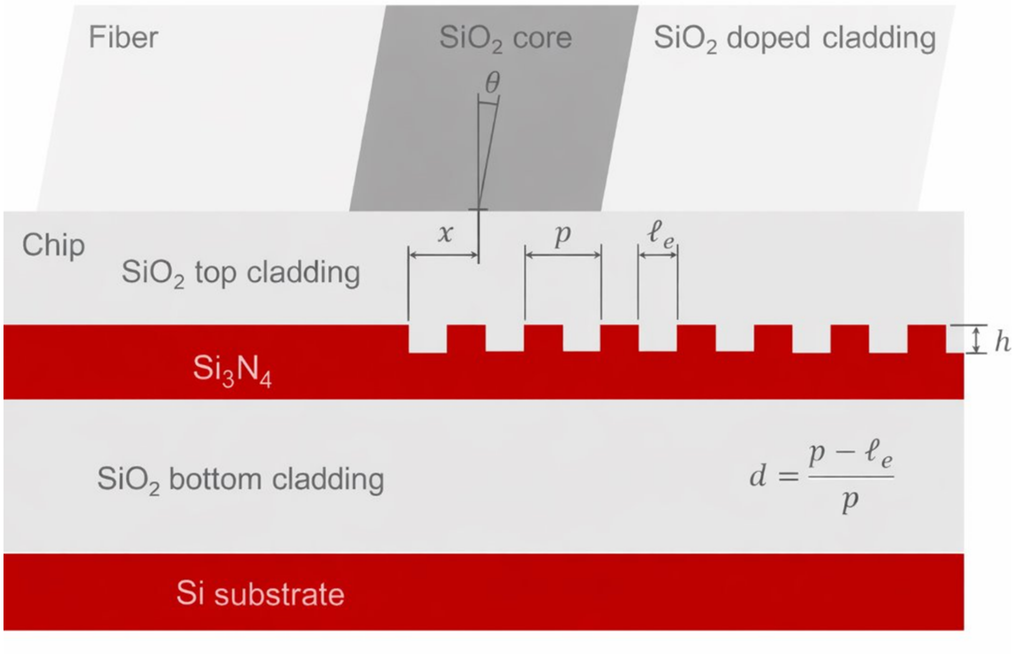


Fig. 4.8. Grating Coupler

4.2.1.) Initial parameters

Initial design parameters were chosen from physical estimates and literature values.

4.2.2.) Set by literature sources

- Fibre angle $= 10°$ [9; 10]
- Core / cladding diameter $= 4.9\ \mu$m / 125 μm [13, p. 4, Table 4]
- Core / cladding refractive index $= 1.44427$ / 1.43482 [14]

4.2.3.) Set by Estimation

$$\Lambda = \frac{m\lambda}{n_{\text{eff}} - n_{\text{clad}} \sin\theta}$$

where:

- m = diffraction order (= 1)
- λ = wavelength being coupled (0.78 μm)
- n_{eff} = effective index of the optical mode (1.649) Calculated using Lumerical MODE [See fig 4.1]
- n_{clad} = effective index of the cladding (1.44)[14]
- θ = fibre angle to the grating normal (10°)[9], [10]

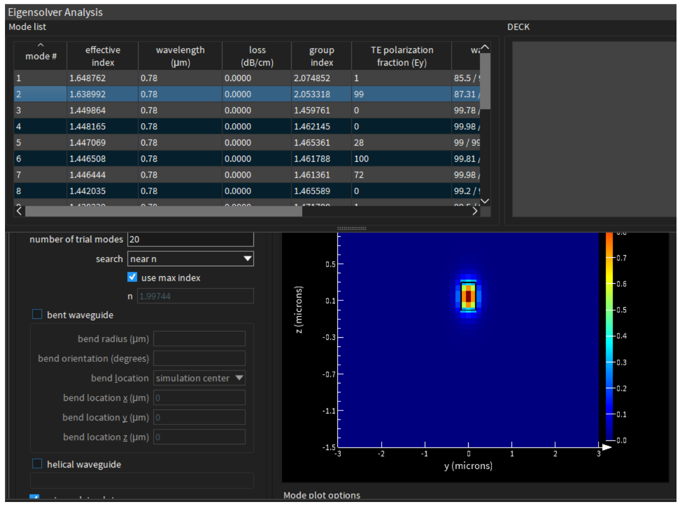


Fig. 4.9. Lumerical FDE

Substituting these values in the grating phase matching equation gives an initial pitch of:

$$\Lambda = 0.558\ \mu\text{m}$$

- Etch depth: initial value 0.15 μm, from a maximum waveguide height of 0.3 μm.
- Duty cycle: initially set to 0.5.
- Grating length: at least 9 μm to accommodate the fibre core; with a 10° fibre tilt,

$$L_{\min} \approx \frac{9}{\cos 10^\circ} \approx 9.14\ \mu\text{m}.$$

4.3.) MMIs

MMIs were selected instead of directional couplers because they offer greater fabrication tolerance and lower sensitivity to dimensional variation [11; 12]. Both 50:50 and 33:66 power splitting were required, so the geometries were optimised for stable splitting ratio and transmission. The main optimisable parameters were:

- MMI core length / width
- input / output taper length / mouth

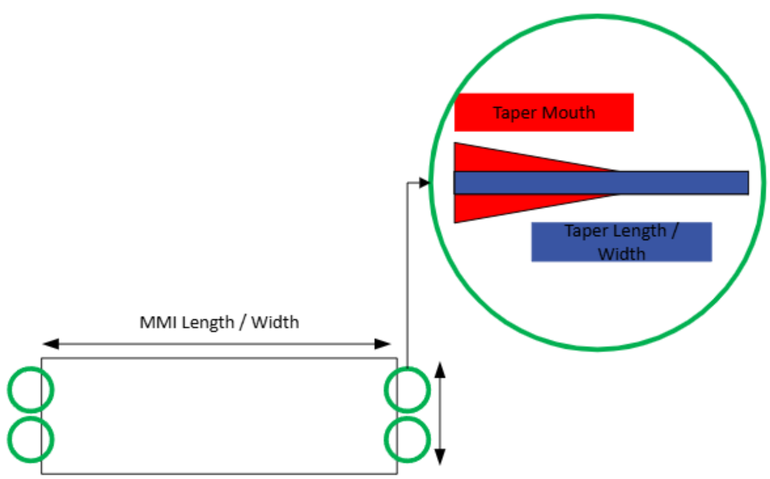


Fig. 4.10. MMI Geometry

the way to measure the components performance is splitting ration and transmissions efficiency.

4.4.) MZIs and AMZI

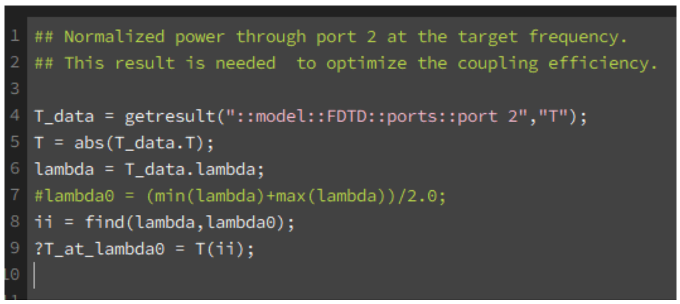
MZIs were used to provide controllable phase shifts within the 4×4 unitary circuit, with thermo-optic tuning chosen for platform compatibility and practical on-chip modulation. AMZIs were used where asymmetric interference was required, particularly for the 1/3 splitting condition.

5. Component Simulation process and Analysis

For grating couplers, FDTD was chosen because it models the full electromagnetic behaviour of the structure, including diffraction, reflection, radiation loss, and fibre-to-waveguide coupling.

5.1.) FOM

Grating-coupler efficiency was measured using the Figure of Merit (FOM), calculated in Lumerical with the script shown in Fig. 5.1.



```
## Normalized power through port 2 at the target frequency.
## This result is needed  to optimize the coupling efficiency.

T_data = getresult("::model::FDTD::ports::port 2","T");
T = abs(T_data.T);
lambda = T_data.lambda;
#lambda0 = (min(lambda)+max(lambda))/2.0;
ii = find(lambda,lambda0);
?T_at_lambda0 = T(ii);
```

Fig. 5.1. FOM Script

5.2.) 2D Simulation

Although frequency-domain monitors were not used to directly measure coupling efficiency, they were useful for visualizing the spatial distribution of optical coupling (Fig. 5.2). A refractive-index monitor was also used to confirm the index contrast affecting the guided-mode effective index, n_{eff} (Fig. 5.3).

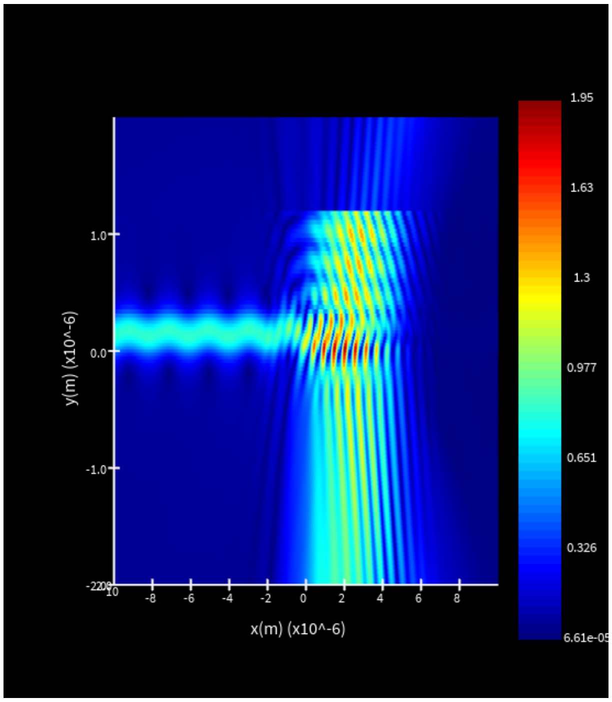


Fig. 5.2. Field-Monitor

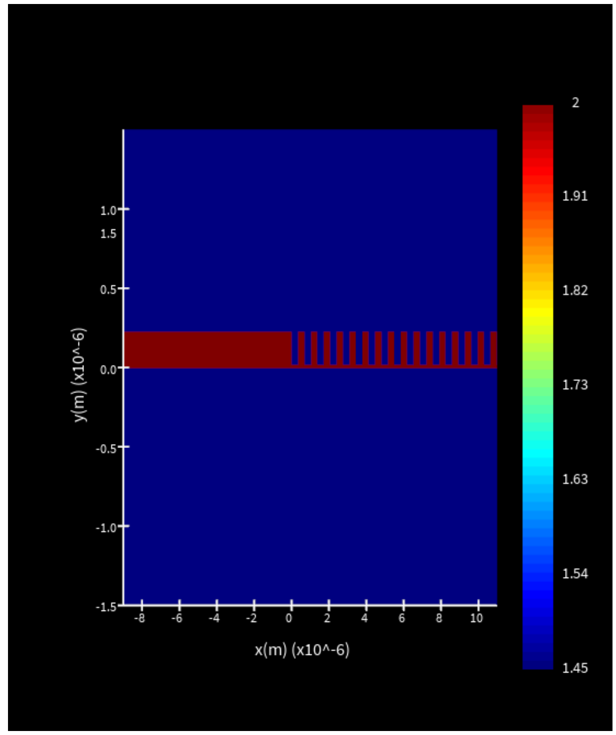


Fig. 5.3. Index-Monitor 1

A distinction was required between parameters varied in 2D and 3D simulations; for example, changing grating width had no effect in 2D due to the limitations of the model (Fig. 5.4).

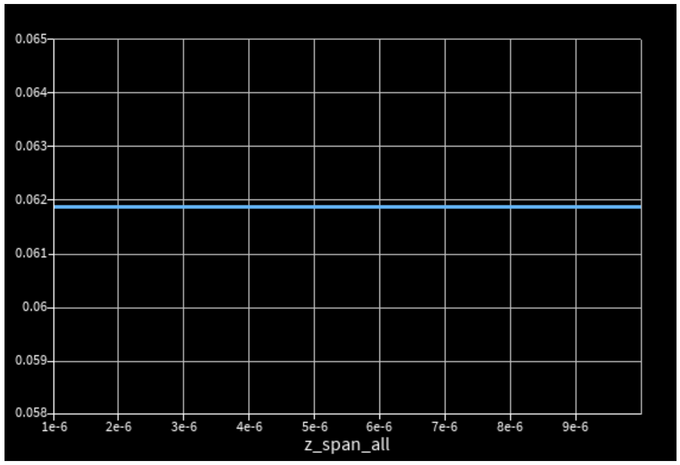


Fig. 5.4. Width Sweep

Table 1. Parameters altered in 2D and 3D simulations

Parameters altered in 2D	Parameters altered in 3D
Pitch	Pitch
Duty cycle	Duty cycle
Etch depth	Etch depth
Fibre angle	Fibre angle
Fibre x-position on grating	Fibre x-position on grating
–	Grating width
–	Grating length

2D parameter sweeps were first performed to identify the approximate optimum region. Figures 5.3–5.6 show the initial pitch and duty cycle sweeps, followed by the re-swept duty cycle and pitch after updating parameter values.

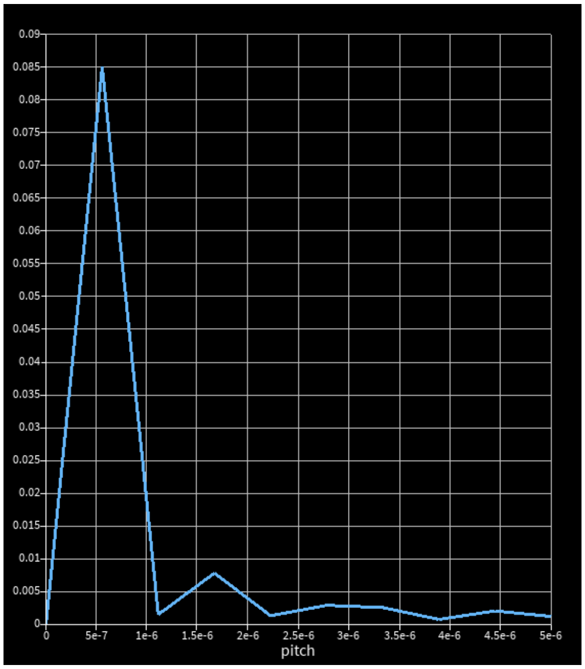


Fig. 5.5. Pitch-Sweep 1

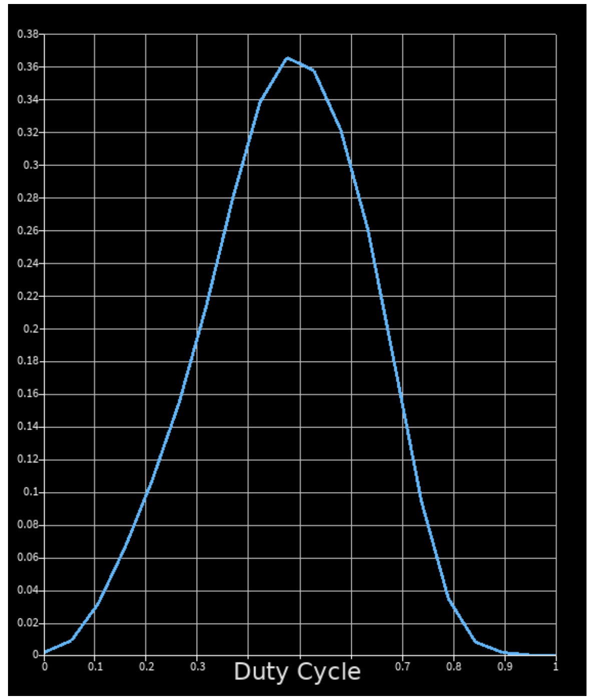


Fig. 5.6. Duty-cycle Sweep 1

The sweeps showed clear parameter interdependence. After updating the pitch, the optimum duty cycle shifted (Figs. 5.6 and 5.7), and re-sweeping the pitch after changing duty cycle showed a further shift in its optimum. This justified the iterative optimization process.

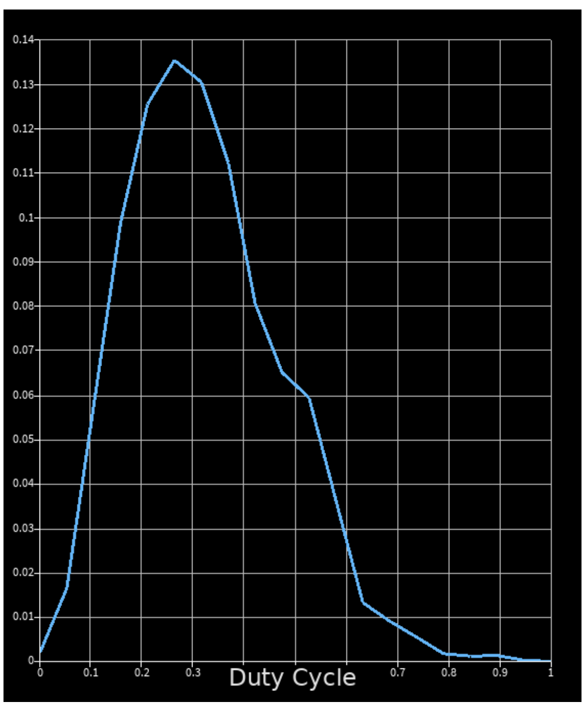


Fig. 5.7. Pitch-Sweep 2

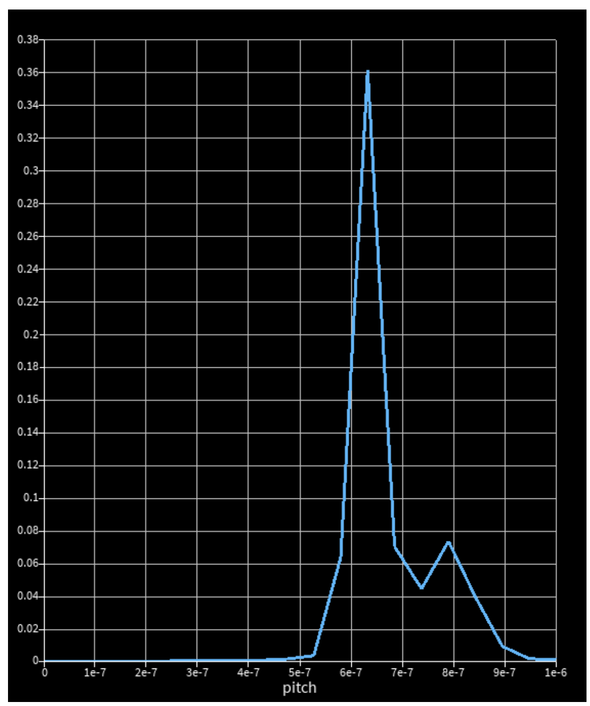


Fig. 5.8. Duty-cycle Sweep 2

This was a crucial factor when developing the full optimization process

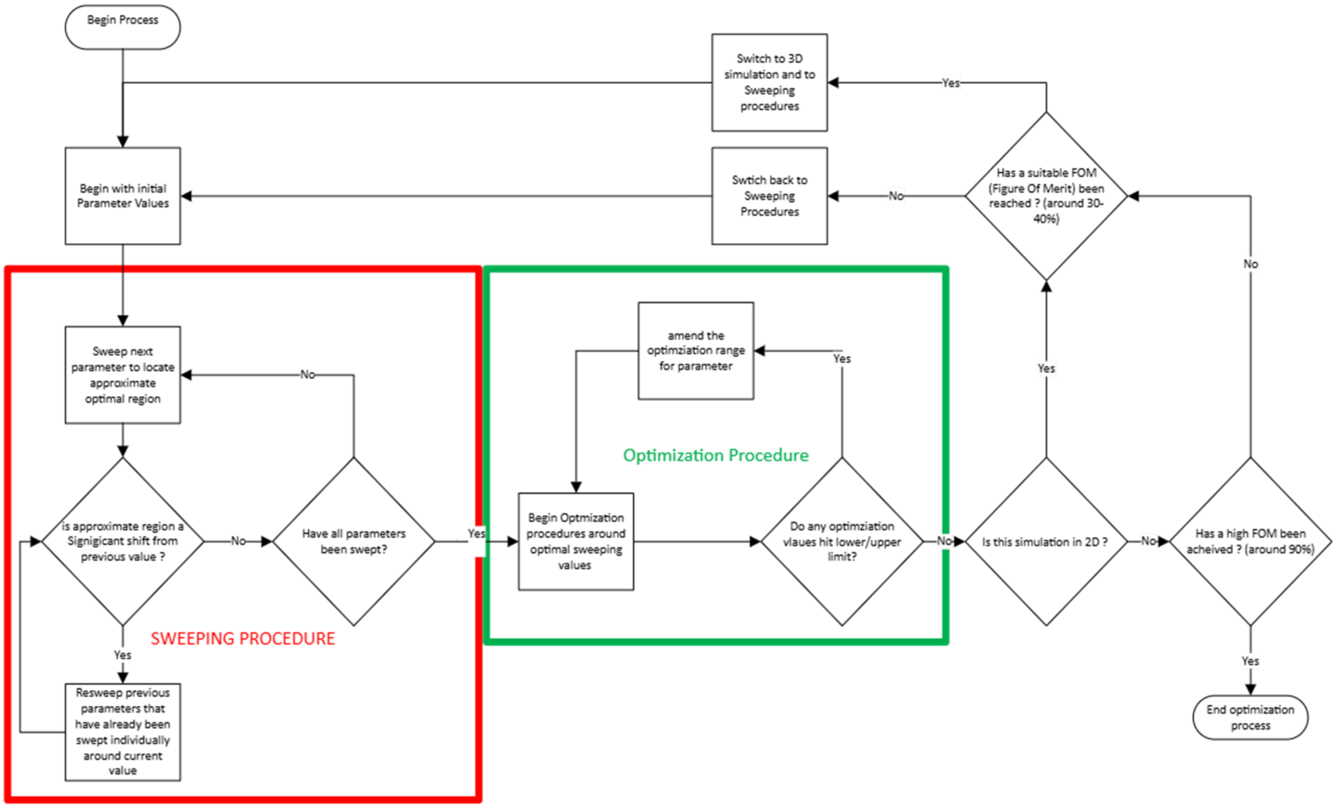


Fig. 5.9. Design-Flow

Table 2. 2D Rectangle Simulation Results

Iteration Number	Pitch (μm)	Etch Depth (μm)	Duty Cycle	Grating Length (μm)	Grating Width (μm)	Fibre Core Position (μm)	Fibre Angle ($^\circ$)	Core Diameter (μm)	Core Index	Cladding Diameter (μm)	Cladding Index	Grating Shape	FOM (%)	Changed Parameter
1	0.558	0.15	0.5	12	6	6	10	4.9	1.44427	125	1.43482	Rectangular	7.28213	Initial
2	0.58	0.15	0.5	12	6	6	10	4.9	1.44427	125	1.43482	Rectangular	8.91	Pitch
3	0.58	0.16	0.5	12	6	6	10	4.9	1.44427	125	1.43482	Rectangular	8.43	Etch Depth
4	0.58	0.16	0.495	12	6	6	10	4.9	1.44427	125	1.43482	Rectangular	9.74	Duty Cycle
5	0.58	0.16	0.495	12	6	6.2	10	4.9	1.44427	125	1.43482	Rectangular	10.52	Fibre Core Position
6	0.58	0.16	0.495	12	6	6.2	11.2	4.9	1.44427	125	1.43482	Rectangular	10.08	Fibre Angle
7	0.595	0.16	0.495	12	6	6.2	11.2	4.9	1.44427	125	1.43482	Rectangular	11.86	Pitch
8	0.6	0.175	0.49	12	6	6.1	12	4.9	1.44427	125	1.43482	Rectangular	13.42	Optimization (all)
9	0.61	0.175	0.49	12	6	6.1	12	4.9	1.44427	125	1.43482	Rectangular	14.65	Pitch
10	0.61	0.185	0.49	12	6	6.1	12	4.9	1.44427	125	1.43482	Rectangular	14.18	Etch Depth
11	0.61	0.185	0.486	12	6	6.1	12	4.9	1.44427	125	1.43482	Rectangular	15.42	Duty Cycle
12	0.61	0.185	0.486	12	6	5.95	12	4.9	1.44427	125	1.43482	Rectangular	16.11	Fibre Core Position
13	0.61	0.185	0.486	12	6	5.95	13.5	4.9	1.44427	125	1.43482	Rectangular	15.73	Fibre Angle
14	0.621	0.185	0.486	12	6	5.95	13.5	4.9	1.44427	125	1.43482	Rectangular	17.26	Pitch
15	0.621	0.198	0.486	12	6	5.95	13.5	4.9	1.44427	125	1.43482	Rectangular	16.88	Etch Depth
16	0.6235	0.2005	0.484114	12	6	5.9	14.8	4.9	1.44427	125	1.43482	Rectangular	19.24	Optimization (all)
17	0.628	0.2005	0.484114	12	6	5.9	14.8	4.9	1.44427	125	1.43482	Rectangular	20.31	Pitch
18	0.628	0.202	0.484114	12	6	5.9	14.8	4.9	1.44427	125	1.43482	Rectangular	19.96	Etch Depth
19	0.628	0.202	0.4825	12	6	5.9	14.8	4.9	1.44427	125	1.43482	Rectangular	21.08	Duty Cycle
20	0.628	0.202	0.4825	12	6	5.85	14.8	4.9	1.44427	125	1.43482	Rectangular	21.76	Fibre Core Position
21	0.628	0.202	0.4825	12	6	5.85	16	4.9	1.44427	125	1.43482	Rectangular	21.33	Fibre Angle
22	0.631579	0.202	0.4825	12	6	5.85	16	4.9	1.44427	125	1.43482	Rectangular	23.04	Pitch
23	0.631579	0.205203	0.4825	12	6	5.85	16	4.9	1.44427	125	1.43482	Rectangular	31.67	Etch Depth
24	0.631579	0.205203	0.4832	12	6	5.8	17.2	4.9	1.44427	125	1.43482	Rectangular	32.88	Optimization (all)
25	0.631579	0.205203	0.484114	12	6	5.8	17.2	4.9	1.44427	125	1.43482	Rectangular	30.42	Duty Cycle
26	0.631579	0.205203	0.484114	12	6	5.8	17.8947	4.9	1.44427	125	1.43482	Rectangular	29.87	Fibre Angle
27	0.601389	0.213067	0.495117	12	6	5.78	15.4248	4.9	1.44427	125	1.43482	Rectangular	35.0043	Optimization (all)

After iterative sweeps and optimization, a suitable parameter set was selected for 3D simulation, since 2D no longer captured all relevant effects.(Fig. 5.10) proves increase coupling.

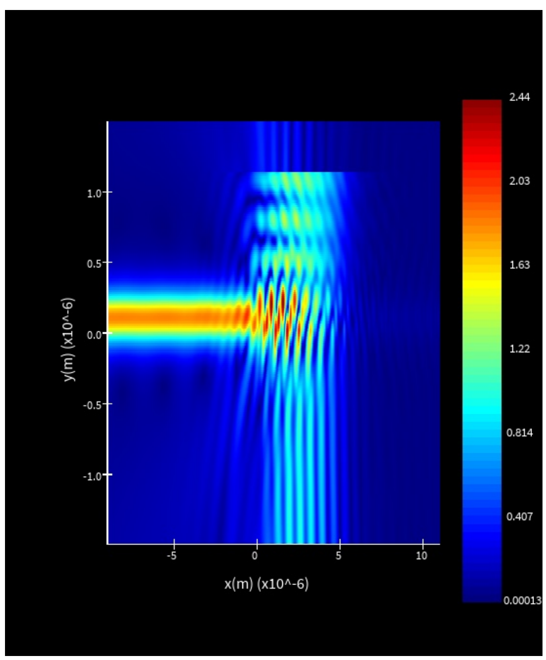


Fig. 5.10. 2D-Field Monitor

5.3.) 3D Simulation

Immediately, the optimal 2D results provided extremely poor efficiency. The other factors that have not been optimized.

Table 3. Inital 3D simulation

Pitch (μm)	Etch Depth (μm)	Duty Cycle	Grating Length (μm)	Grating Width (μm)	Fibre Core Position (μm)	Fibre Angle (°)	Core Diameter (μm)	Core Index	Cladding Diameter (μm)	Cladding Index	Grating Shape	FOM (%)
0.601389	0.213067	0.495117	12	6	5.78	15.4248	4.9	1.44427	125	1.43482	Rectangular	0.0000109749

5.3.1.) Coupling in Fibre

To model coupling from a tilted fibre in 3D, the port orientation and rotation offset were calculated from the fibre angle. First, the fibre angle was converted from degrees to radians:

$$\theta_{\text{fiber,rad}} = \theta_{\text{fiber}} \left(\frac{\pi}{180} \right) \quad [1]$$

The port angle was then defined as

$$\theta_{\text{port}} = -\theta_{\text{fiber}} \quad [2]$$

so that the injected field propagated along the tilted fibre axis. A rotation offset was also introduced to account for the lateral displacement caused by the fibre inclination [15]:

$$\Delta x = 4D_{\text{core}} \tan\left(\theta_{\text{fiber,rad}}\right) \quad [3]$$

where Δx is the port rotation offset and D_{core} is the fibre core diameter. This correction improved alignment between the excitation port and the tilted fibre, helping to produce a more accurate representation of fibre-to-grating coupling in the 3D simulation. Fig (5.11) verifies good coupling

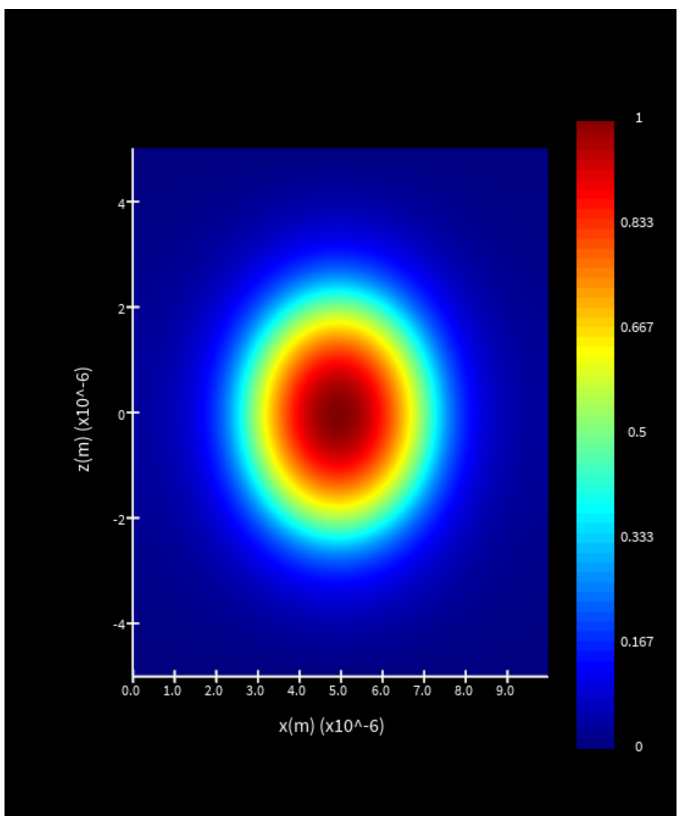


Fig. 5.11. Fibre Mode

5.3.2.) Grating Width limitation

Grating width had limited influence once it was large enough to overlap the fibre core, so further optimisation was unnecessary (fig 5.12).

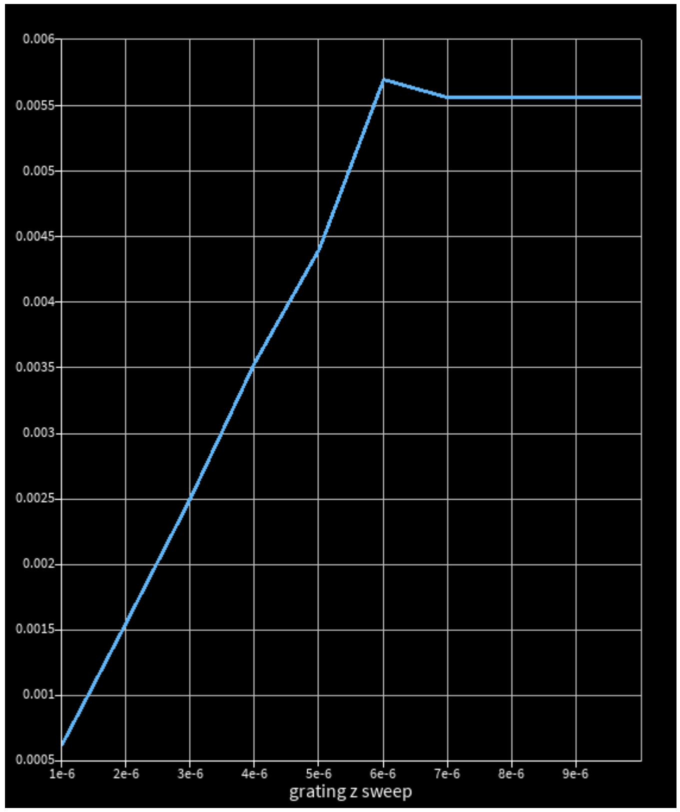


Fig. 5.12. Width-Sweep1

5.3.3.) 3D Results

Table 3 outlines the de-satisfying values, providing the hint that more drastic changes were required

Table 4. 3D Rectangle Simulation Results

3D Iteration Number	Pitch (μm)	Etch Depth (μm)	Duty Cycle (0–1)	Grating Length (μm)	Grating Width (μm)	Fibre Core Position (μm from waveguide)	Fibre Angle (°)	Core Diameter (μm)	Core Refractive Index	Cladding Diameter (μm)	Cladding Refractive Index	Grating Shape	FOM (%)	Changed Parameter
1	0.601389	0.213067	0.495117	12	6	6	15.4248	4.9	1.44427	125	1.43482	Rectangle	1.09749E-05	Initial
2	0.59	0.213067	0.495117	12	6	6	15.4248	4.9	1.44427	125	1.43482	Rectangle	0.00012	Pitch
3	0.59	0.205	0.495117	12	6	6	15.4248	4.9	1.44427	125	1.43482	Rectangle	0.0001	Etch Depth
4	0.59	0.205	0.492	12	6	6	15.4248	4.9	1.44427	125	1.43482	Rectangle	0.00019	Duty Cycle
5	0.59	0.205	0.492	11.7	6	6	15.4248	4.9	1.44427	125	1.43482	Rectangle	0.00023	Grating Length
6	0.59	0.205	0.492	11.7	6	5.2	15.4248	4.9	1.44427	125	1.43482	Rectangle	0.00021	Fibre Core Position
7	0.59	0.205	0.492	11.7	6	5.2	14.2	4.9	1.44427	125	1.43482	Rectangle	0.00031	Fibre Angle
8	0.582	0.198	0.4895	11.2	6	4.6	13.2	4.9	1.44427	125	1.43482	Rectangle	0.00048	Optimizing
9	0.575	0.198	0.4895	11.2	6	4.6	13.2	4.9	1.44427	125	1.43482	Rectangle	0.00044	Pitch
10	0.575	0.19	0.4895	11.2	6	4.6	13.2	4.9	1.44427	125	1.43482	Rectangle	0.00059	Etch Depth
11	0.575	0.19	0.4865	11.2	6	4.6	13.2	4.9	1.44427	125	1.43482	Rectangle	0.00073	Duty Cycle
12	0.575	0.19	0.4865	10.7	6	4.6	13.2	4.9	1.44427	125	1.43482	Rectangle	0.00067	Grating Length
13	0.575	0.19	0.4865	10.7	6	3.8	13.2	4.9	1.44427	125	1.43482	Rectangle	0.00086	Fibre Core Position
14	0.575	0.19	0.4865	10.7	6	3.8	12.1	4.9	1.44427	125	1.43482	Rectangle	0.00081	Fibre Angle
15	0.575	0.19	0.4865	10.1	6	3.8	12.1	4.9	1.44427	125	1.43482	Rectangle	0.00102	Grating Length
16	0.565	0.18	0.484	9.5	6	2.7	11	4.9	1.44427	125	1.43482	Rectangle	0.00129	Optimizing
17	0.558	0.18	0.484	9.5	6	2.7	11	4.9	1.44427	125	1.43482	Rectangle	0.00118	Pitch
18	0.558	0.17	0.484	9.5	6	2.7	11	4.9	1.44427	125	1.43482	Rectangle	0.00147	Etch Depth
19	0.558	0.17	0.4815	9.5	6	2.7	11	4.9	1.44427	125	1.43482	Rectangle	0.00166	Duty Cycle
20	0.558	0.17	0.4815	9	6	2.7	11	4.9	1.44427	125	1.43482	Rectangle	0.00155	Grating Length
21	0.558	0.17	0.4815	9	6	1.2	11	4.9	1.44427	125	1.43482	Rectangle	0.00198	Fibre Core Position
22	0.558	0.17	0.4815	9	6	1.2	10	4.9	1.44427	125	1.43482	Rectangle	0.00186	Fibre Angle
23	0.558	0.17	0.4815	8.6	6	1.2	10	4.9	1.44427	125	1.43482	Rectangle	0.00212	Grating Length
24	0.548	0.16	0.479	8.1	6	0.15	9.2	4.9	1.44427	125	1.43482	Rectangle	0.00241	Optimizing
25	0.54	0.16	0.479	8.1	6	0.15	9.2	4.9	1.44427	125	1.43482	Rectangle	0.00229	Pitch
26	0.54	0.15	0.479	8.1	6	0.15	9.2	4.9	1.44427	125	1.43482	Rectangle	0.00253	Etch Depth
27	0.54	0.15	0.4765	8.1	6	0.15	9.2	4.9	1.44427	125	1.43482	Rectangle	0.00272	Duty Cycle
28	0.54	0.15	0.4765	7.9	6	0.15	9.2	4.9	1.44427	125	1.43482	Rectangle	0.00261	Grating Length
29	0.54	0.15	0.4765	7.9	6	0.01	9.2	4.9	1.44427	125	1.43482	Rectangle	0.00298	Fibre Core Position
30	0.54	0.15	0.4765	7.9	6	0.01	8.95	4.9	1.44427	125	1.43482	Rectangle	0.00285	Fibre Angle
31	0.54	0.15	0.4765	7.6	6	0.01	8.95	4.9	1.44427	125	1.43482	Rectangle	0.00311	Grating Length
32	0.528	0.135	0.474	7.4	6	0.00001	8.7	4.9	1.44427	125	1.43482	Rectangle	0.00356	Optimizing
33	0.522	0.135	0.474	7.4	6	0.00001	8.7	4.9	1.44427	125	1.43482	Rectangle	0.00338	Pitch
34	0.522	0.128	0.474	7.4	6	0.00001	8.7	4.9	1.44427	125	1.43482	Rectangle	0.00385	Etch Depth
35	0.522	0.128	0.4728	7.4	6	0.00001	8.7	4.9	1.44427	125	1.43482	Rectangle	0.00422	Duty Cycle
36	0.522	0.128	0.4728	7.3	6	4.03024E-06	8.20049	4.9	1.44427	125	1.43482	Rectangle	0.00274478	Grating Length
37	0.522	0.128	0.4728	7.2	6	4.54622E-06	7.41282	4.9	1.44427	125	1.43482	Rectangle	0.002	Fibre Core Position
38	0.522	0.128	0.4728	7.1	6	4.04045E-06	8.5	4.9	1.44427	125	1.43482	Rectangle	0.00254392	Fibre Angle
39	0.522	0.128	0.4728	7	6	3.1592E-06	5	4.9	1.44427	125	1.43482	Rectangle	0.016623	Grating Length
40	0.511408	0.119062	0.4712	6.9	6	4.18654E-06	8.95381	4.9	1.44427	125	1.43482	Rectangle	0.0181321	Optimizing

6. Geometry Amendment

The rectangular geometry was not supported by any literature sources, instead a more curved / coned shape was common with efficient GC [16], [17]. All data since now has been from GC-V1 in Fig 6.2

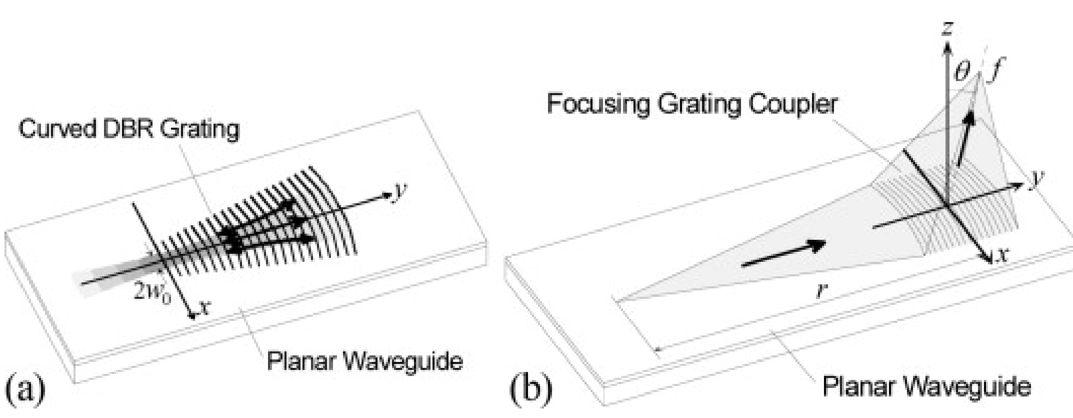


Fig. 6.1. Coned-GC [18]

Fig. 6.2. GC-V1

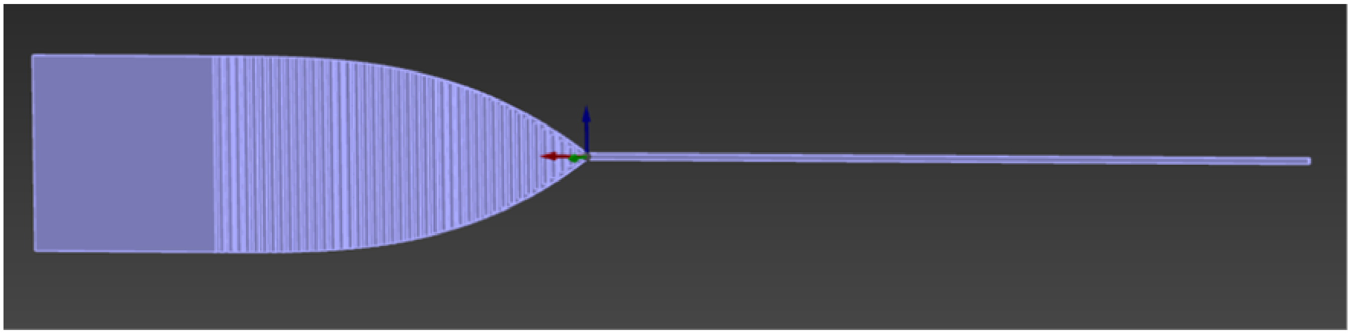


Fig. 6.3. GC-V2

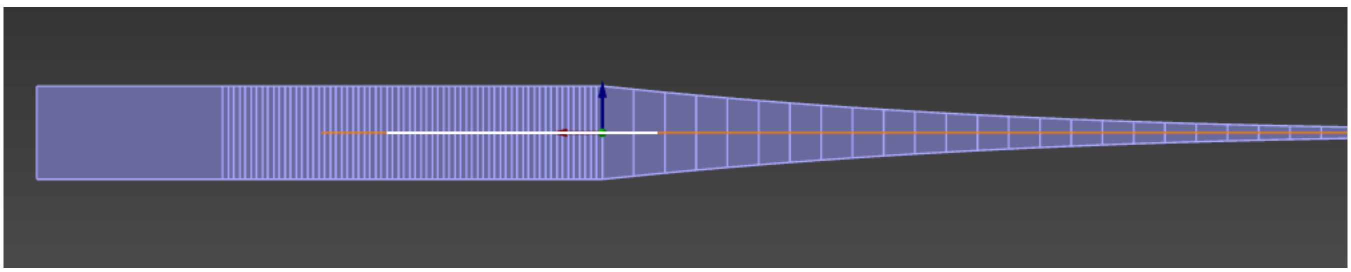


Fig. 6.4. GC-V3

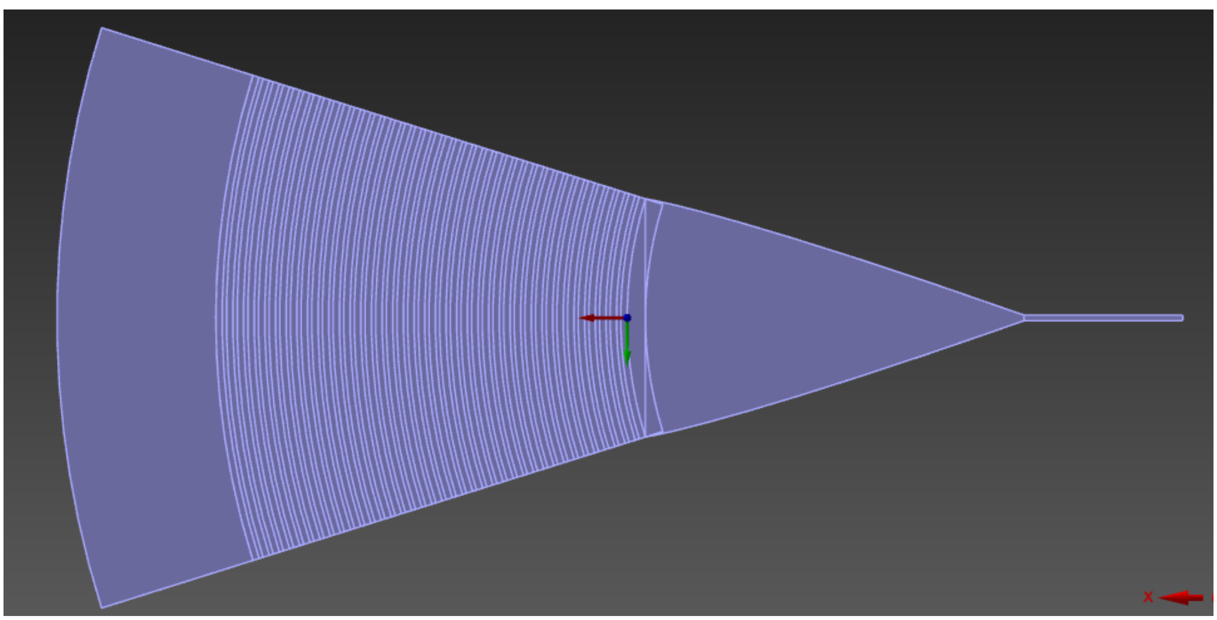


Fig. 6.5. GC-V4

As discussed in Section 5.2, increasing grating width alone did not improve performance. GC-V2 (Fig. 6.3) performed significantly worse, with increased scattering causing severe coupling loss (Fig. 6.6), likely due to poorer matching to the narrow TE waveguide mode.

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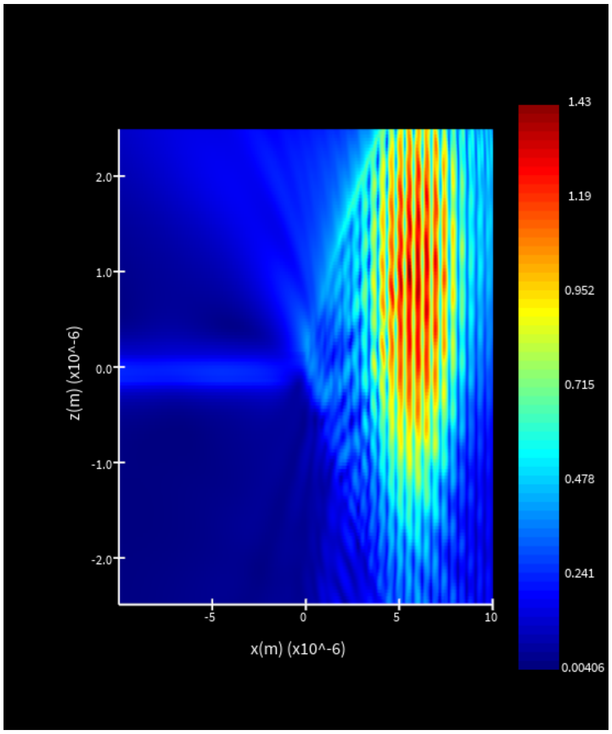

Fig. 6.6. Field-Monitor1

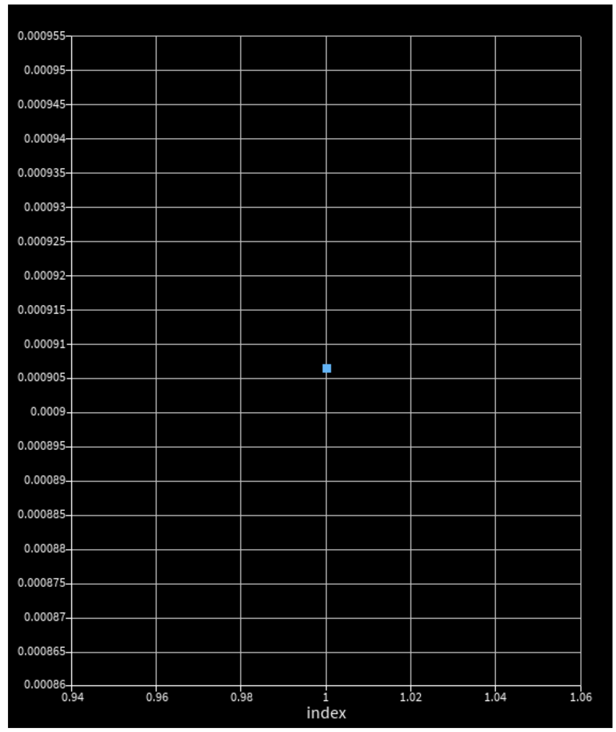

Fig. 6.7. Index-Monitor2

6.1.) Taper

A taper was necessary for smoother transmission from grating region to waveguide. to test the optimal taper parameters, it was simulated separately, and used a horizontal light source rather than a coupled vertical source

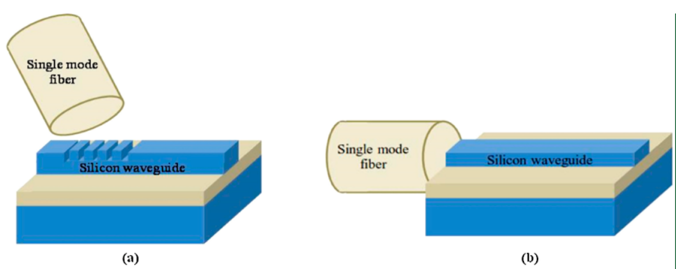

Fig. 6.8. Taper Light Source [19]

To calculate the taper angle [α], use the following equation:

$$\alpha = \tan^{-1}\left(\frac{W_g - W_{wg}}{2L_t}\right)$$

- W_g = grating width (6um)
- W_{wg} = waveguide width (0.35um)
- L_t = taper length
- α = taper half-angle

In general, a lower [α] leads to a smoother transition, therefore greater coupling efficiency.

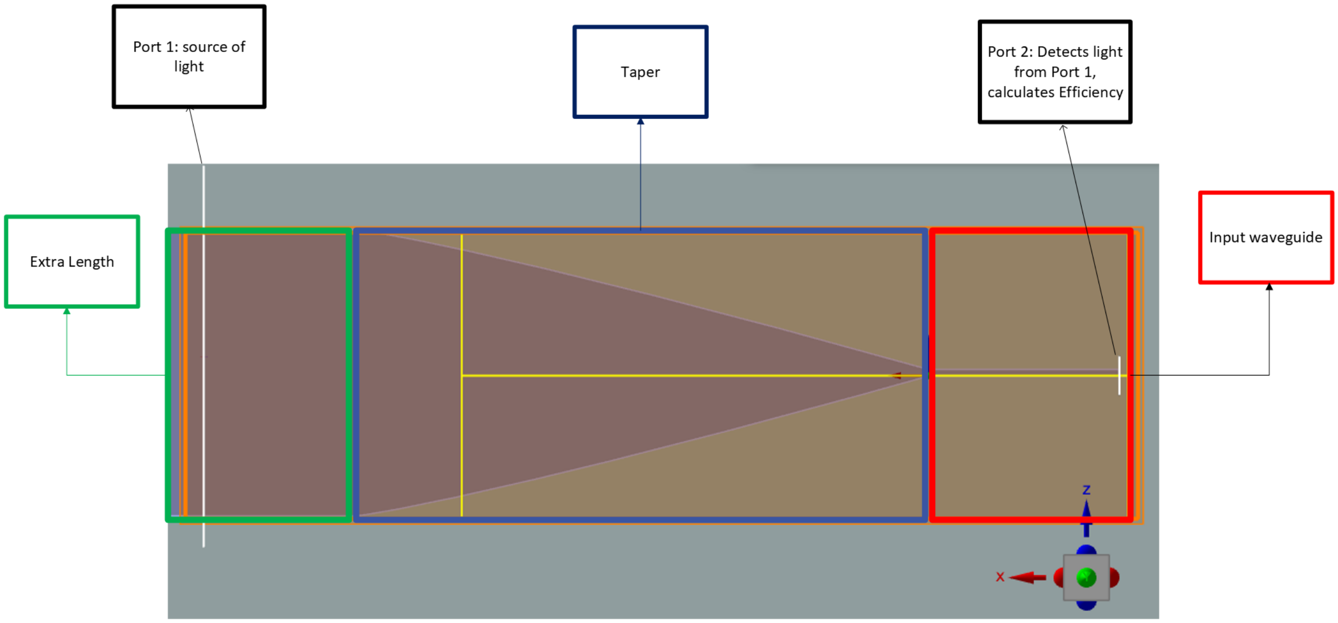

Fig. 6.9. Taper-Model (From Fig 6.8)

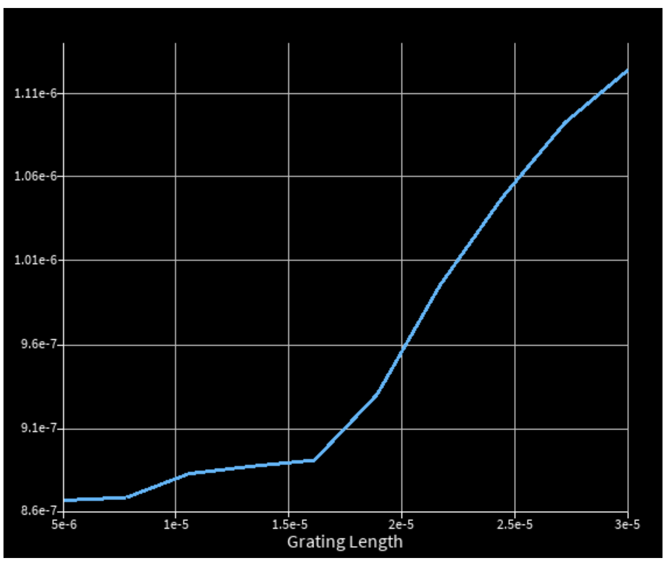

Fig. 6.10. Taper-Length Sweep

Taper length, L_t (μm)	Angle, α ($^\circ$)	FOM (%)
5	29.47	0.866
7.77	19.98	0.868
10.55	14.99	0.882
13.33	11.97	0.887
16.11	9.95	0.89
18.88	8.51	0.93
21.66	7.43	0.995
24.44	6.59	1.05
27.22	5.93	1.09
30	5.38	1.12

Table 5. Taper length, taper half-angle, and FOM.

The graph insists we have not find the optimal area yet due to its strong increase at the upper limit. Ranges had to be increased (Table 6)

Taper length, L_t (μm)	Angle, α ($^\circ$)	FOM
25	0	
50	25	
75	50	
100	75	0.000000508203

Table 6. Taper length, taper half-angle, and FOM.

6.2.) Curved Gratings

For the curved design, the grating opening angle was estimated from simple circular geometry, $\phi = 2\sin^{-1}(W_g/(2R))$. Changes in figure of merit were then used to relate geometric modifications directly to coupling performance.

$$\phi = 2\sin^{-1}\left(\frac{W_g}{2R}\right)$$

- W_g = grating width (um)
- R = curvature radius of grating teeth (Degree)
- ϕ = grating opening angle (degree)

Creating the Coned-shaped grating couplers we see in literature sources mentioned in section 6 – add table of data

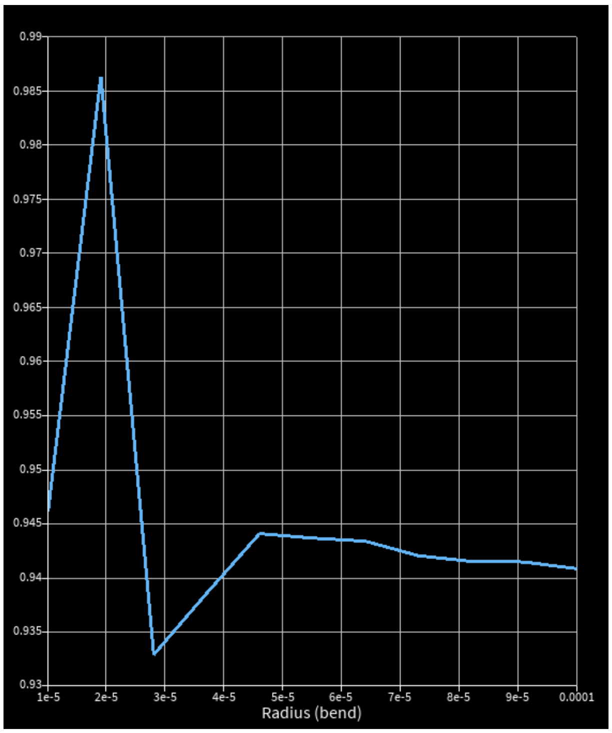

Fig. 6.11. Radius-Sweep 1

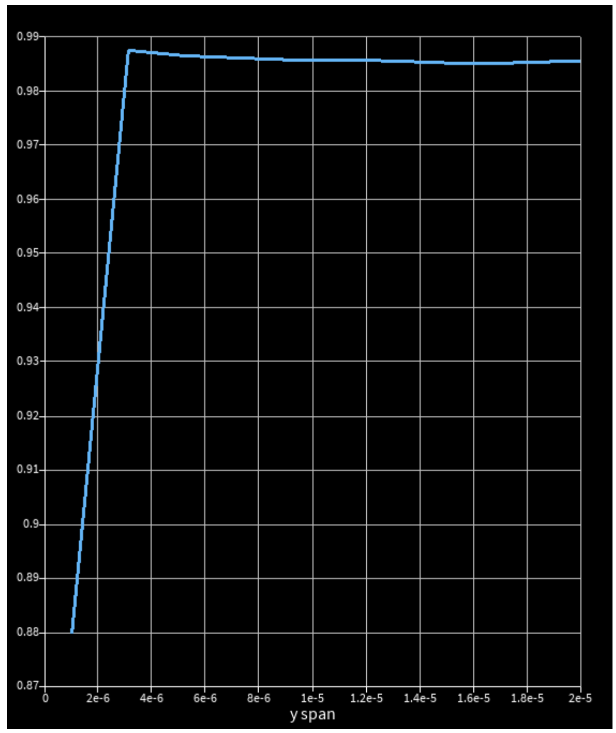

Fig. 6.12. Grating-Width 1

Grating Width W_g	Radius Sweep R	Grating opening angle ϕ	FOM (%)
15	10	Undefined	94.4
15	25	34.92	95.1
15	50	17.25	94.4
15	75	11.48	94.2
15	100	8.60	94.1

Table 7. Radius Sweep FOM correlation

Grating Width W_g	Radius Sweep R	Grating opening angle ϕ	FOM (%)
1	25	2.29	87.8
3.11	25	7.13	98.7
9.44	25	21.80	98.5
13	25	30.08	98.5
20	25	47.16	98.6

Table 8. Grating width FOM correlation.

A parameter sweep was performed over bounds informed by previous data (Tables 6 and 7) to optimise both parameters, justified by their strong interdependence (Figs. 6.13–6.14).

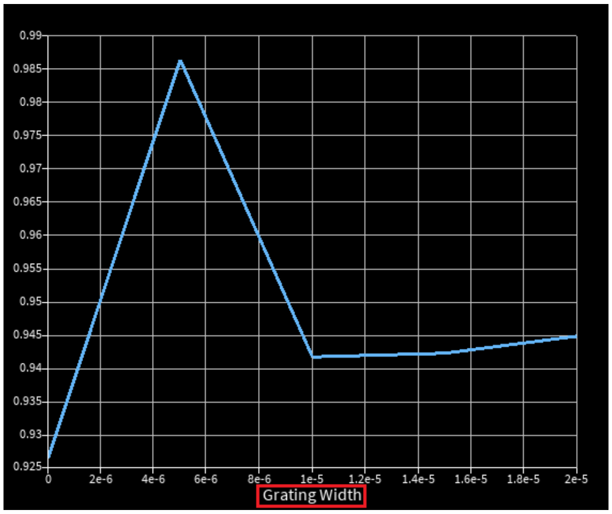

Fig. 6.13. Grating-Width 2

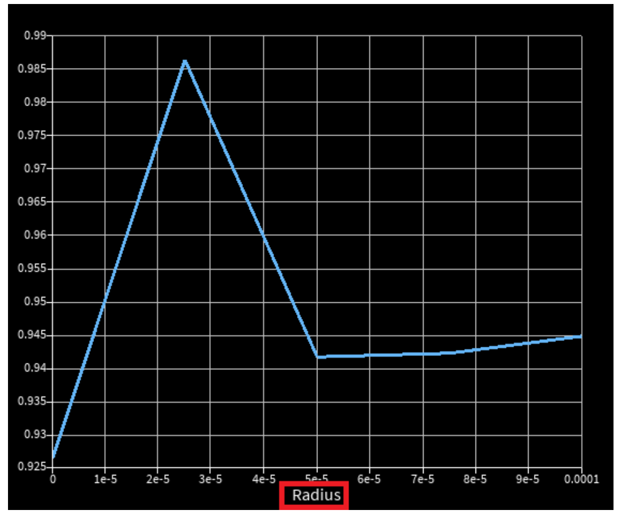

Fig. 6.14. Radius-Sweep 2

7. GDS Assembly

The final grating coupler achieved a FOM of 98.5274%, with the optimized design parameters shown in Table 9.

Table 9. Final grating coupler design parameters

Pitch (μm)	Etch Depth (μm)	Grating Length (μm)	Taper Length (μm)	Taper Half-Angle ($^\circ$)	Duty Cycle	Period Number	Radius ($^\circ$)	Taper Width (μm)	Extra Waveguide Length (μm)	Grating Width (μm)
0.6713	0.1	25	10	14.99	0.3992	38	25	15	10	5

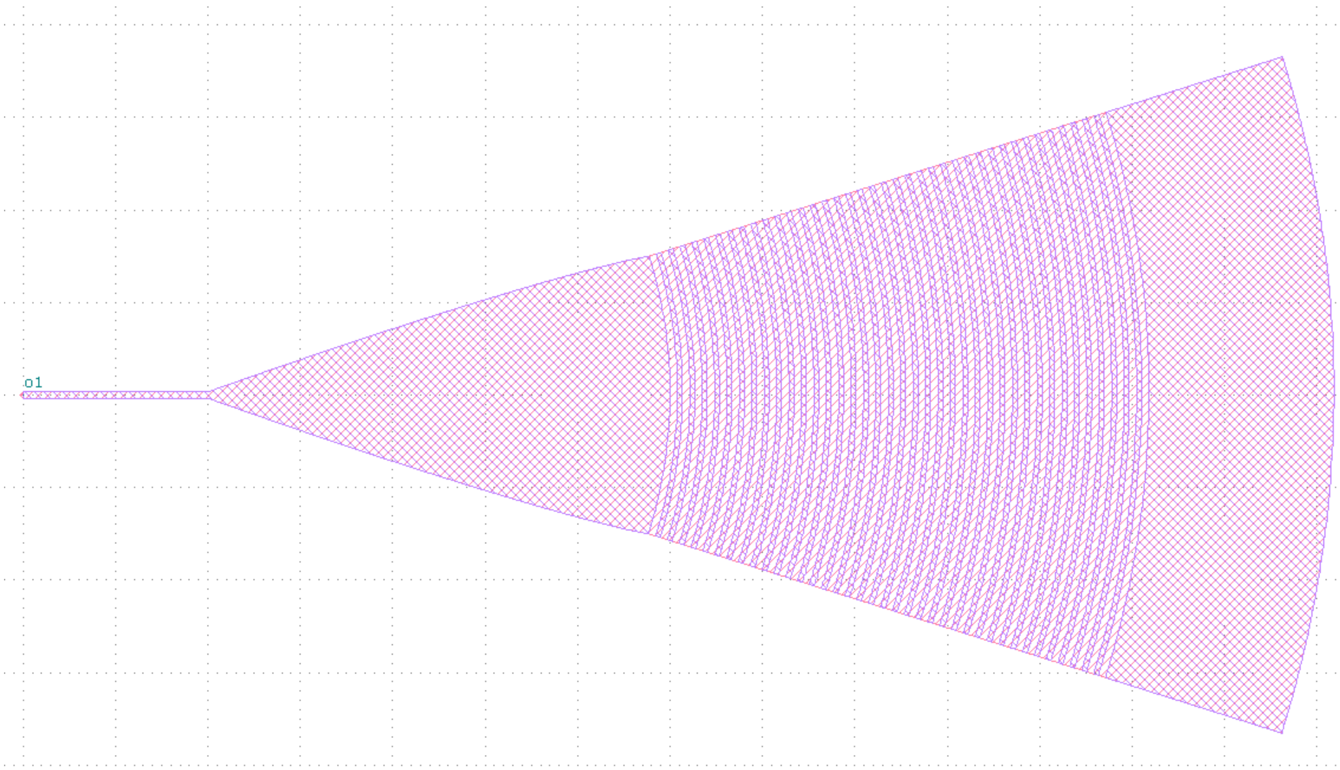

Fig. 7.1. GC GDS

All other components were exported into GDS files

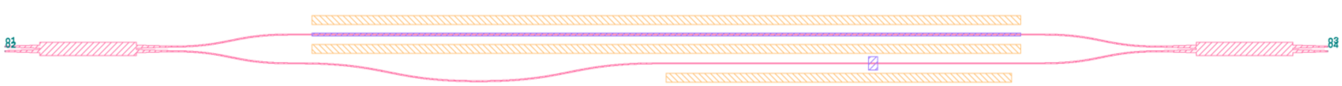

Fig. 7.2. 33:66 Coupler/AMZI

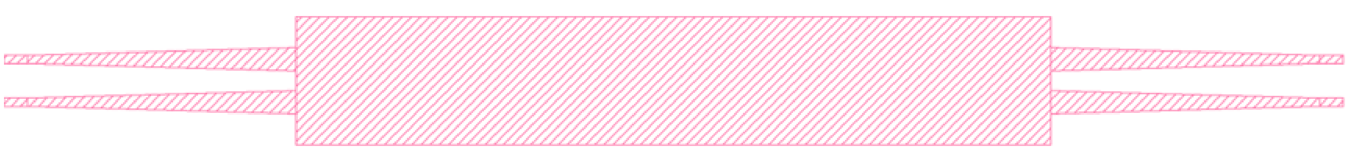

Fig. 7.3. 50:50 MMI

Fig. 7.4. MZI

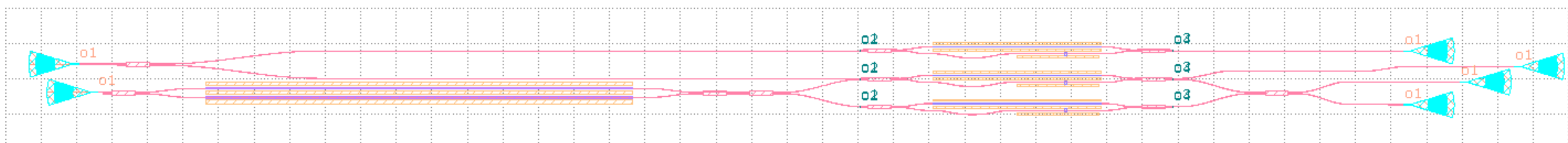

Fig. 7.5. Full-GDS

8. Conclusion

In conclusion, the project demonstrated that a proof-of-concept SiN photonic CNOT architecture at 780 nm is feasible with strong simulated component performance.

- rating coupler achieved 98.53(%) efficiency (section 7)
- the optimised 50:50 MMI reached 96.6(%) transmission at 32.25 m core length
- and the MZI retained an estimated 93.32(%) efficiency

AMZI also achieved the required asymmetric splitting, with a ratio of 0.333 and an overall transmission of about 78.6(%).

Table 10. AMZI simulated S-bend geometries

S-bend height, y (μm)	Ratio
5.91	0.31692
5.8905	0.33067
5.9	0.33316
5.8895	0.34761
5.89	0.40811
5.889	0.42098
5.88	0.49421

AMZI also achieved the required asymmetric splitting, with a ratio of 0.333 and an overall transmission of about 78.6(%). The optimized 50:50 MMI reached 96.6(%) transmission at 32.25um core length, and the MZI retained an estimated 93.32(%) efficiency. The AMZI also achieved the required asymmetric splitting, with a ratio of 0.333 and an overall transmission of about 78.6(%).

$$\eta_{\text{path}} = \eta_{\text{GC}}^2 \eta_{\text{MMI}} \eta_{\text{MZI}} \eta_{\text{AMZI}} = (0.9853)^2 (0.966)(0.9332)(0.786) \approx 0.6879 \approx 68.8\%$$

Combined, these results suggest that a representative cascaded optical path would preserve about 68.8(%) of the input power, showing that the design provides a solid basis for future refinement, fabrication, and full quantum validation.

Total word count: 1488

9. References

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